

SUPERMARKET SALES REVENUE ANALYSIS

Domain Executive Briefing: Retail Domain (supermarket_sales.csv) | Date: August 2026

1. Executive Summary

Time-series and revenue analysis of supermarket transactions assessing daily revenue momentum, tax contribution, and hourly shopping footfall.

2. Key Performance Indicators (KPI Overview)

Metric Description	Key Performance Value	Strategic Benchmark / Context
Total Gross Revenue	■623,843.99	Total across all transactions
Average Invoice Total	■311.92	Average spend per customer
Top Revenue City	Naypyitaw	Highest revenue location

3. Formal Statistical Hypothesis Testing

Customer Spend Analysis: --- FORMAL HYPOTHESIS TEST --- H■: Members and Normal customers spend the same amount on average. H■: Members spend more than Normal customers. Test used: Independent Two-Sample T-Test Test statistic (t-stat): 0.3294 p-value: 0.74190 95% Confidence Interval: [-16.8882, 23.706] Effect Size (Cohen's d): 0.0147 Significance level: 0.05 Decision: Fail to Reject H■ Business interpretation: There is no statistically significant difference in average transaction value between Member and Normal customers. Why this matters: The result does not provide sufficient statistical evidence for a targeted intervention. -----

4. Core Analytical & Statistical Insights

- Insight 1:** Finding: Daily revenue exhibits periodic volume spikes that coincide with weekends. Meaning: Customers tend to prefer weekend shopping trips.
- Insight 2:** Finding: No significant difference in member vs normal spend. Meaning: The loyalty program is not currently associated with higher transaction values.
- Insight 3:** Finding: Peak activity is 4-7 PM. Meaning: After-work shopping is associated with increased volume.

5. Graphical Analytics & Visual Evidence

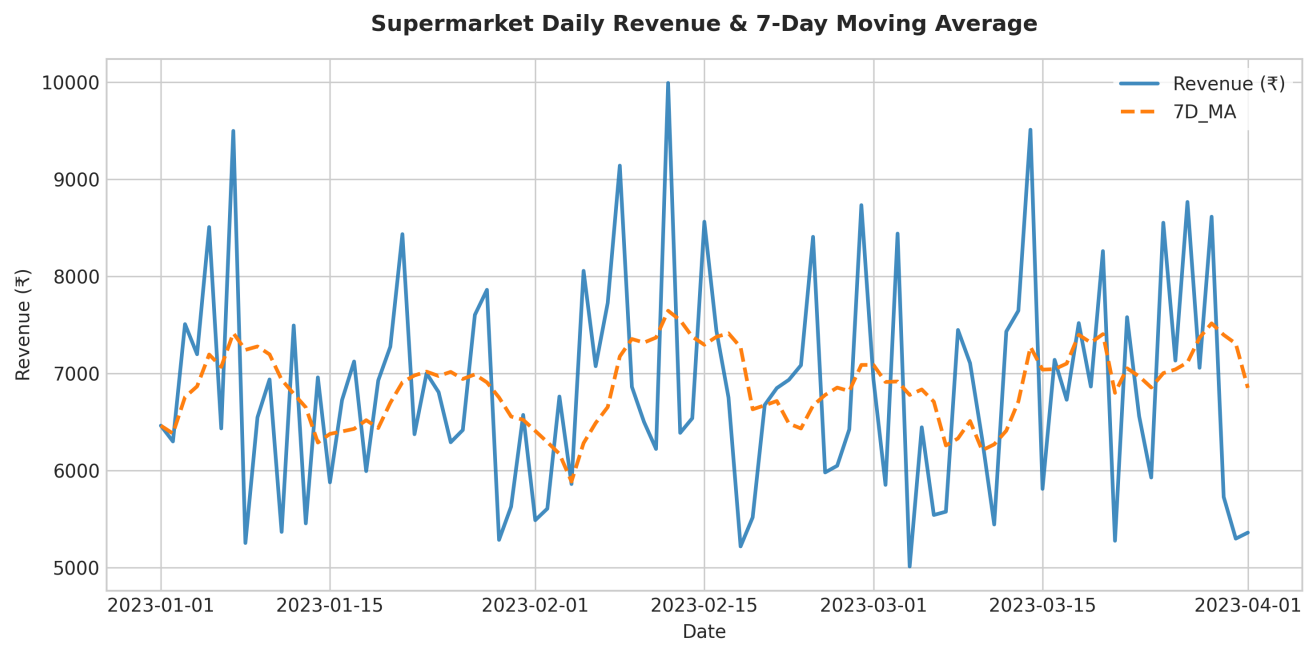


Figure 1: Visual analytical representation derived from Retail Domain (supermarket_sales.csv) dataset.

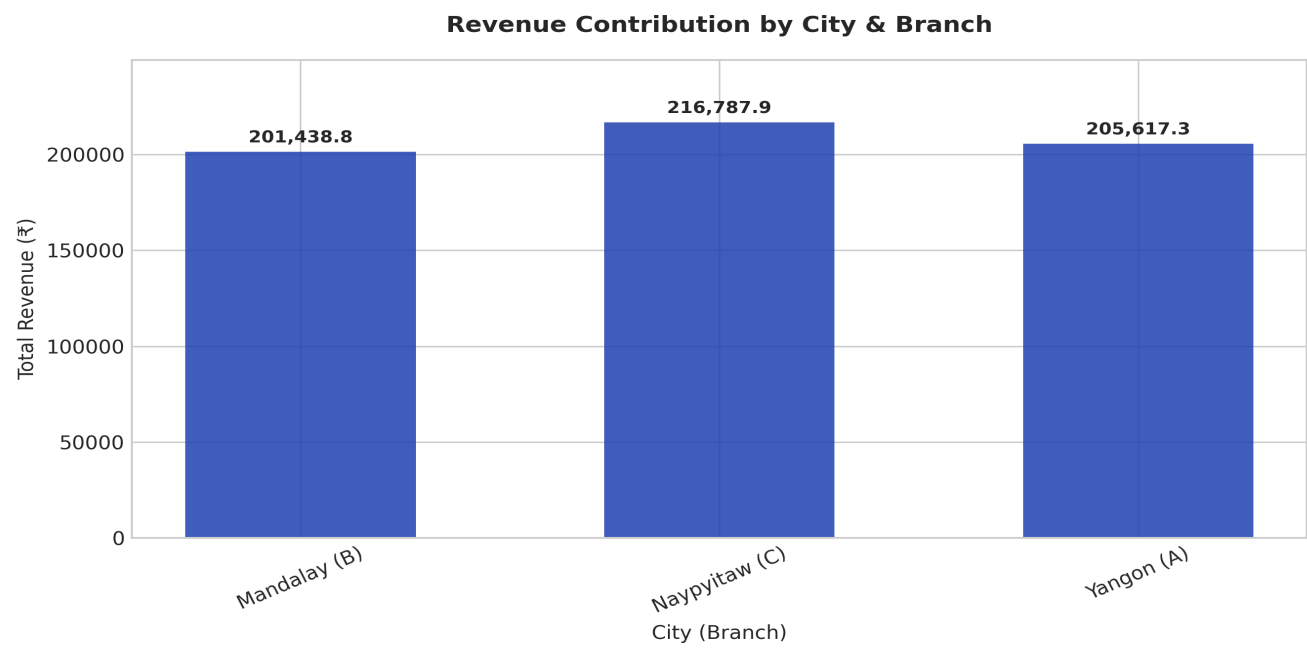


Figure 2: Visual analytical representation derived from Retail Domain (supermarket_sales.csv) dataset.

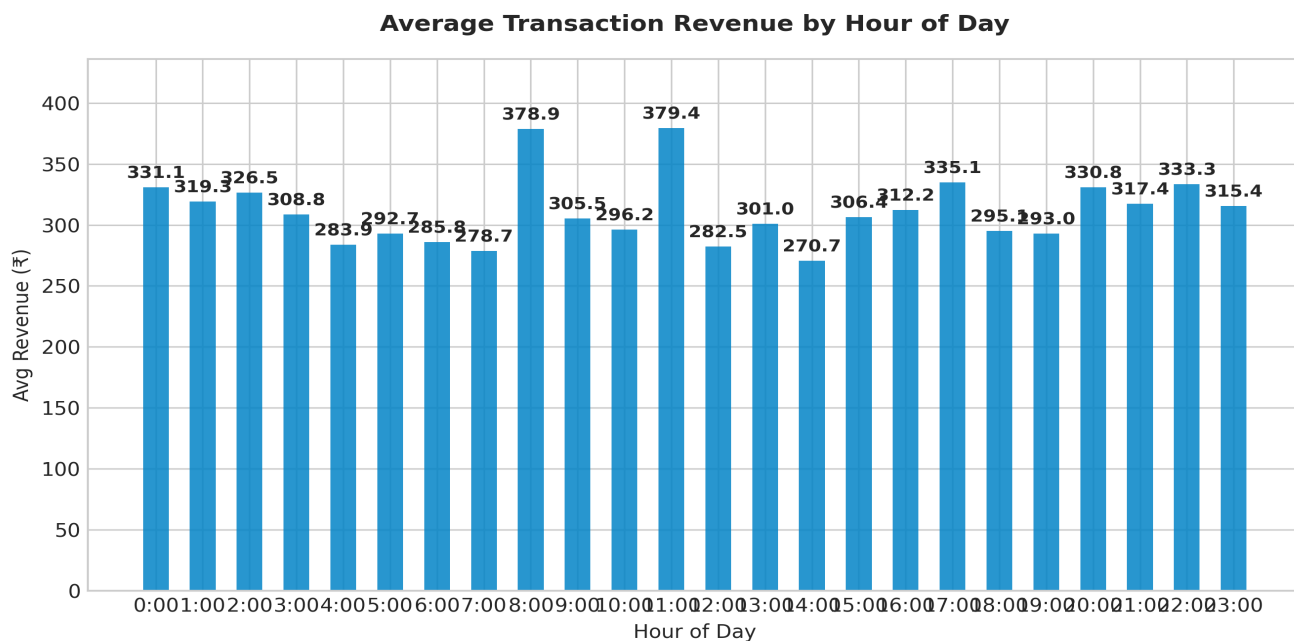


Figure 3: Visual analytical representation derived from Retail Domain (supermarket_sales.csv) dataset.

6. Strategic Recommendations & Action Plan

Action 1: Consider targeted weekend promotional campaigns to maximize existing footfall.

Action 2: Evaluate scheduling extra checkout staff between 4:00 PM and 7:00 PM to reduce queue times.